Shell-Coded Casimir Boundary Transduction: Physical Readout Architectures for the White Sphere–Cylinder Geometry

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Abstract

White et al. introduced a custom Casimir geometry in which a micron-scale sphere inside a conducting cylinder produced a shell-like worldline-numerics vacuum response. A role-resolved follow-on audit separated that result into morphology, calibrated scale, measurement-channel participation, and ordinary material/electromagnetic backgrounds. The durable target is a shell-forming finite-boundary response: the annular sphere–wall region remains the physical feature to interrogate, central timing is an amplitude-bounded reference, and material response is calibration.

The continuation is a boundary-transduction test: a geometry-locked Casimir response measured through ordinary precision apparatus channels with explicit material and electromagnetic controls. Here a readout is the complete measurement ensemble: transducer, observable, control geometry, and nuisance templates. Three readout architectures survive branch triage: a paired differential-pressure cell for direct stress, a high-Q electromagnetic cavity for eigenmode perturbation, and a superconducting resonator for cryogenic microwave boundary response.

A shell-coherent finding would identify a repeatable boundary-conditioned vacuum response in a finite shell-forming geometry. Background-selected and sensitivity-bounded outcomes would map how real machines turn geometry, materials, patches, resonator loss, and cryogenic conditions into measurable signals.

1. INTRODUCTION

Casimir experiments translate boundary-conditioned quantum fields into measurable apparatus behavior. Conductors, dielectrics, superconductors, and structured finite geometries alter the electromagnetic mode spectrum. The renormalized stress associated with that altered spectrum appears in mechanical force, pressure, force gradients, resonant frequency shifts, phase shifts, quality-factor changes, and temperature-dependent superconducting responses. Contemporary Casimir metrology is therefore an instrumentation discipline as much as a field-theory calculation: it depends on surface potential control, separation metrology, material characterization, roughness models, finite-conductivity corrections, resonator stability, and environmental rejection [4, 5, 6, 13, 9, 11, 14, 15].

White, Vera, Han, Bruccoleri, and MacArthur used worldline numerics to study custom Casimir cavities and reported a sphere-in-cylinder case whose negative energy-density morphology resembles the shell placement drawn in an Alcubierre-style source section [1, 2]. The compact case uses a 1 μm diameter sphere centered in a nominal 4 μm diameter cylinder. The geometry separates the stress-shaping body, inner wall, annular gap, material boundary, exterior control region, and possible measurement channel into distinct experimental roles.

White et al.'s proposed detection concept centered on voltage and current response in custom cavities and, for the sphere–cylinder toy model, a central current, photon, or electron transit comparison against a control. The present readout families ask which complete apparatus ensemble can carry a geometry-locked shell coefficient after material, electrostatic, thermal, resonator, and loss templates are fitted. The measurement schedule modulates the shell coordinate, records material and electromagnetic backgrounds, and fits the shell template against nuisance templates. Branch triage assigns central timing to a reference channel, material response to calibration, and pressure, cavity, and superconducting readouts to the physical continuation.

The three apparatus families are supported by existing machinery: MEMS/NEMS Casimir force sensors [6, 11], high-Q nanomembrane probes with Kelvin-probe surface-potential control [9, 10], long-range differential force measurements with explicit patch and roughness accounting [13], and superconducting Casimir or microwave-resonator platforms [12, 16, 17, 18].

2. THE PHYSICAL CLAIM

The shell-coded response concerns renormalized stress-energy in a finite boundary value problem. Ford's review gives the QFT status of negative energy densities and the quantum inequality restrictions that accompany them [3]. Here the measurable content is the geometry-indexed projection of boundary-conditioned stress into force, pressure, energy, or mode behavior.

Finite geometries add spatial structure: a boundary can concentrate the response in a specific region, and an apparatus can couple to that region with varying strength. The White sphere–cylinder geometry supplies a candidate spatial code whose annular sphere–wall region acts as the shell role. The readout test measures how that role couples to a pressure cell, a cavity mode, or a superconducting resonator with enough strength and template separation for comparison against ordinary material and electromagnetic controls.

3. RELATION TO WHITE ET AL.

White et al. supplied a geometry, a worldline calculation, and a provocative metric comparison [1]. The present analysis separates the pieces needed for experimental use:

1. Geometry: a small, finite, role-structured sphere–cylinder boundary.
2. Morphology: a shell-like organization of the vacuum-boundary response.
3. Scale: the calibrated stress level associated with that response.
4. Transduction: the physical apparatus channel that converts the response into measured data.

The first two pieces specify the controlled geometry: the shell band is the region to modulate, and the body-wall spacing is the coordinate that moves the shell template. The calibrated scale piece assigns direct central timing and direct metric readout to an amplitude-bounded reference role. Material and impedance observables calibrate ordinary capacitance, contact, conductivity, and loss responses. Differential pressure carries direct stress; high-Q cavity and superconducting readouts carry resonant frequency, phase, mode-splitting, or microwave response with nuisance templates in the same fit.

3.1 Three Physical Projections of the Boundary Response

The three surviving branches are complementary projections of the same finite-boundary response. The White-derived geometry specifies a boundary-conditioned stress organization in the annular sphere–wall region; the apparatus reads it as mechanical stress, electromagnetic eigenmode perturbation, or cryogenic microwave boundary response.

The pressure projection belongs to the classical Casimir measurement lineage. The ideal parallel-plate result supplies the clean stress formula, and the experimental history moved quickly into torsion balances, sphere–plane measurements, AFM probes, and MEMS/NEMS sensors because controlled micron-scale parallelism is hard [4, 5, 6, 11]. Patch potentials, roughness, finite conductivity, optical data, and electrostatic calibration are core measurement variables [13, 14, 15]. The pressure cell extends that lineage from a standard plate/sphere force law to a finite shell-coded stress contrast.

The cavity projection belongs to resonant metrology. Slater-style cavity perturbation reads small metallic or dielectric perturbations as frequency shifts whose size depends on local field overlap [7]. Modern cavity optomechanics generalizes the same instrumental idea: small boundary, material, or mechanical changes are converted into frequency, phase, linewidth, and mode-coupling observables [8]. In this branch the shell-coded boundary response is the perturbation source, and the selected cavity eigenmode is the transducer.

The superconducting projection is a cryogenic boundary-metrology problem. Superconductors alter electromagnetic boundary response, but real microwave devices are dominated by kinetic inductance, quasiparticles, vortices, dielectric loss, residual surface resistance, magnetic history, drive power, and thermal cycling [16]. Recent experiments now reach direct Casimir-superconductivity tests, including condensation-energy concepts, superconducting drum optomechanics, and high-resolution force sensing across a superconducting transition [12, 17, 18]. The branch amplifies small effects alongside highly structured material channels.

The same geometry schedule is therefore tested against three independent nuisance ledgers.

4. READOUT MODEL

Let q denote a controlled geometry or apparatus setting: gap, sphere radius or surrogate radius, cylinder radius, wall thickness, lateral offset, cell length, mode overlap, readout radius, temperature, magnetic history, or drive power. The measured observable $y(q)$ is modeled as

$$y(q) = a_{\text{shell}} T_{\text{shell}}(q) + \sum_i a_i T_i(q) + \varepsilon(q), \tag{1}$$

where T_{shell} is the shell-coded response template and the T_i are measured background templates. The background family includes patch potential, capacitance, roughness, membrane drift, temperature, vibration, finite conductivity, waveguide dispersion, mode mixing, kinetic inductance, surface loss, vortex response, quasiparticles, and material controls.

Evidence is selection of a_{shell} over the nuisance family under material-only and electromagnetic-only controls. The output is a coefficient ledger: shell coefficient, nuisance coefficients, uncertainties, and false-positive checks. A geometry-locked response vector separates shell, material, and readout behavior. Table 1 lists the resulting measurement families.

The design-screening ladder associated with this paper used a signal/background recovery gate of $\mathrm{S/B} = 1.2$ and an electromagnetic-only false-positive target near $\mathrm{FP} < 0.05$. Table 2 summarizes the portable design implications.

Table 1: Physical readout families for the shell-coded boundary response.

Family	Observable	Physical transduction	Main control role
Differential pressure	Paired pressure or force contrast	Boundary stress in the shell-gap geometry	Patch, roughness, membrane, thermal, and vibration rejection
High-Q cavity	$\delta f/f$, phase, Q, or mode splitting	Electromagnetic eigenmode perturbation from shell-sensitive boundary overlap	Loss, thermal, waveguide, and mode-mixing separation
Superconducting resonator	Microwave $\delta f/f$, phase, Q, nonlinear drum response, or T_c-coupled response	Cryogenic boundary response in a superconducting microwave apparatus	Surface loss, kinetic inductance, vortex, quasiparticle, and material separation
Material impedance	Capacitance, impedance, contact, or loss response	Ordinary material and electromagnetic apparatus behavior	Calibration map for backgrounds

Table 2: Screening-level design constraints for the physical branches.

Branch	Representative viable cases	Predicted S/B range	Design implication
High-Q cavity	nominal clean through optimistic clean/matched/lossy	2.22–19.34	Strongest design row; mode overlap and loss templates set interpretation
Superconducting resonator	nominal balanced/clean through optimistic clean/balanced/lossy	3.59–17.76	Strong cryogenic path; surface-loss and quasiparticle controls are central
Differential pressure	optimistic balanced and optimistic quiet	2.02–5.31	Direct force path; patch and common-mode controls set feasibility
Material impedance	calibration rows	background map	Measures ordinary apparatus response for the same geometry

5. COMMON APPARATUS ARCHITECTURE

All three readout branches share an experimental skeleton: a shell-sensitive cell, a matched dummy cell, a material-control channel, a geometry actuator, surface metrology, and a monitored readout chain. The shell cell implements the sphere–cylinder geometry or a manufacturable surrogate. The dummy and material channels preserve ordinary response while suppressing shell overlap.

The common setup begins with branch-native calibration: a known force or pressure law, a known cavity perturbation or optomechanical response, or the ordinary temperature, magnetic-history, drive-power, and loss maps of a superconducting resonator. Those calibrations make the White-geometry schedule evidence-bearing.

The geometry schedule is part of the measurement. Gap, radius, offset, wall thickness, length, and readout overlap move the shell template differently from patch, capacitance, material, and drift templates. Surface measurements accompany those settings. Garcia-Sanchez et al. integrated Kelvin-probe surface-potential mapping into high-Q nanomembrane Casimir measurements [9]. Bimonte et al. demonstrate a long-separation force measurement with explicit treatment of patch potentials, roughness, edge effects, uncertainty, and theory comparison [13]. Patch-specific modeling and Kelvin-probe measurements are part of the apparatus definition [14, 15].

Valid runs use matched dummy and material channels to separate shell selection from geometry-correlated apparatus response.

6. DIFFERENTIAL-PRESSURE ARCHITECTURE

Force and pressure are the original operational face of Casimir energy. The parallel-plate calculation gives the simplest stress law, and high-precision experiments largely adopted sphere–plane and microdevice geometries because large, perfectly parallel micron gaps are difficult to build and hold [4, 5, 6]. The resulting literature made surface preparation, electrostatic calibration, finite conductivity, roughness, patch potentials, and separation metrology central to the physics result [13, 14, 15]. The White-derived pressure cell extends that lineage into a finite shell-coded stress contrast.

The differential-pressure architecture reads boundary stress directly. Two matched cells share environment and readout while their geometry settings produce different shell-gap responses. The primary observable is

$$\Delta P(q) = P_{\text{shell}}(q) - P_{\text{control}}(q), \tag{2}$$

with the fit model of Eq. 1. For ideal parallel plates,

$$P_{\text{C}}(a) = -\frac{\pi^2 \hbar c}{240 a^4}, \tag{3}$$

which gives about 1.3×10^{-3} Pa at $a = 1\,\mu$m. The sphere–cylinder cell is finite and curved, so Eq. 3 serves as a scale reference for the apparatus model. The screening rows place the strongest pressure cases in the same metrological neighborhood, with shell pressures around 10^{-4}–10^{-3} Pa.

6.1 Expected Pressure Signatures

A White-positive pressure observation would be a geometry-locked differential stress vector. The shell cell and dummy cell would share environment, membrane mechanics, material stack, wiring, and readout chain. Opposite shell-gap modulation would produce a pressure contrast following the annular sphere–wall shell template. Gap, radius, wall, offset, and length settings would move the fitted shell coefficient in the White-derived schedule. Large-gap, far-field, and non-shell controls would suppress that coefficient. Kelvin-probe maps and roughness measurements would account for electrostatic and surface terms while leaving a residual shell-scheduled pressure component.

Alternative selections would assign the response to measured apparatus channels: patch selection to surface-potential maps, roughness or finite conductivity to surface and optical priors, membrane or vibration selection to mechanical drift, and electrostatic calibration to capacitance and bias schedules. A sensitivity-bounded outcome would place the shell and nuisance coefficients below the inherited signal/background recovery gate.

6.2 Pressure Controls and Screening

Established Casimir force platforms define the required toolchain: calibrated actuation, separation sensing, patch mapping, roughness measurement, and common-mode rejection [6, 11, 9, 10].

In the priority physical screen, pressure survives in two strong-control optimistic rows: S/B = 5.31 for the quiet case and S/B = 2.02 for the balanced case. The required branch model combines paired-cell Lifshitz pressure, patch-potential priors, roughness, membrane stress, and common-mode rejection.

7. HIGH-Q CAVITY ARCHITECTURE

The high-Q cavity branch belongs to the cavity perturbation and resonant metrology tradition. In a resonator, a small boundary, dielectric, conductor, or mechanical perturbation becomes a shift in frequency, phase, linewidth, or mode splitting. Maier and Slater's resonant-cavity perturbation work already used small inserted objects to infer local electromagnetic field components from frequency shifts [7]. Cavity optomechanics later made the same transduction logic a broad precision-measurement platform in which mechanical displacement and boundary motion modulate optical or microwave eigenmodes [8]. The White shell geometry therefore enters this branch as a boundary perturbation source whose usefulness is determined by mode overlap.

The high-Q cavity architecture uses resonance as the transducer. A cavity mode with large field overlap in the shell-sensitive region converts a small boundary perturbation into a measurable frequency, phase, Q, or mode-splitting signal. In first-order perturbation language,

$$\frac{\delta f}{f} \sim -\frac{1}{2}\frac{\int_\Omega \left(\delta\epsilon\,|E|^2 - \delta\mu\,|H|^2\right)\,dV}{\int_\Omega \left(\epsilon|E|^2 + \mu|H|^2\right)\,dV} + \Delta_{\text{boundary}}, \tag{4}$$

where Δ_{boundary} denotes the shell-sensitive finite boundary contribution modeled by the apparatus eigenmode solver. The equation states the design requirement: maximize known shell overlap and measure the ordinary material terms in the same basis. The central theoretical variables are shell-mode overlap and nuisance separation: the shell-coded boundary response projects onto a chosen eigenmode with a schedule distinct from thermal, loss, waveguide, and mode-mixing channels in a successful readout.

7.1 Expected Cavity Signatures

A White-positive cavity observation would be a mode-indexed shell coefficient. A high-shell-overlap mode would show a frequency, phase, Q, or mode-splitting response following the White shell schedule. A low-shell-overlap mode, null mode, or dummy cavity would suppress the coefficient. The eigenmode solver would predict the sign and relative magnitude trend across gap, radius, wall, offset, and mode-overlap settings. Thermal expansion, conductor loss, waveguide, port coupling, and material-wall templates would remain separately fitted coordinates.

Alternative selections would assign the response to branch-native apparatus channels: mode mixing to avoided crossings, thermal response to expansion monitors, conductor loss to surface and optical priors, waveguide or port coupling to readout-chain sweeps, and material response to the dummy cavity.

A sensitivity-bounded outcome would indicate limited shell overlap or elevated resonator noise in the tested mode.

The screening ladder ranks this branch highly. The strongest cavity row gives S/B = 19.34, and the nominal clean row remains above the recovery gate. Those values require a shell-overlap mode, a null mode or dummy cavity, thermal metrology, port-coupling sweeps, and a validated eigenmode model.

8. SUPERCONDUCTING-RESONATOR ARCHITECTURE

The superconducting branch sits at the intersection of superconducting microwave resonators, Casimir forces involving superconductors, condensation energy, and cryogenic optomechanics. Superconducting boundaries change the electromagnetic response of a cavity, but practical resonator behavior is often set by kinetic inductance, quasiparticles, trapped vortices, dielectric participation, residual surface resistance, drive-power dependence, and thermal history [16]. Recent Casimir-superconductivity experiments make the branch experimentally timely, from NEMS transition-temperature concepts to superconducting drum force readout and on-chip force sensing across a superconducting transition [12, 17, 18]. The test is superconducting-boundary metrology: the shell-coded Casimir geometry is tested inside the loss-channel decomposition of a real cryogenic microwave device.

The superconducting architecture uses a cryogenic microwave resonator, drum, or a cavity sensitive to the superconducting transition to read the shell-coded response. The readout observable is frequency shift, phase shift, Q change, nonlinear mechanical response, or a superconducting transition response. The core fit is

$$\left(\frac{\delta f}{f}\right)_{\mathrm{sc}} = a_{\mathrm{shell}}T_{\mathrm{shell}} + a_{\mathrm{ki}}T_{\mathrm{ki}} + a_{\mathrm{surf}}T_{\mathrm{surf}} + a_{\mathrm{vort}}T_{\mathrm{vort}} + a_{\mathrm{qp}}T_{\mathrm{qp}} + a_{\mathrm{mat}}T_{\mathrm{mat}} + \varepsilon, \tag{5}$$

with kinetic inductance, surface loss, vortex loading, quasiparticles, and material response treated as measured templates.

8.1 Expected Superconducting Signatures

A White-positive superconducting observation would be a microwave response locked to the shell schedule after cryogenic material terms are measured. A shell-coupled resonator would differ from a material-only resonator on the same cooldown path. Low-shell-overlap modes, dummy resonators, or non-shell geometries would suppress the coefficient. Temperature sweeps below and across T_{c}, magnetic-history tests, vortex-loading controls, drive-power sweeps, and quasiparticle checks would leave a stable shell coefficient whose geometry dependence follows the sphere–wall schedule.

Alternative selections would assign the response to cryogenic material channels: kinetic inductance to temperature, current, and film participation; quasiparticles to population and drive history; vortices to magnetic preparation and cooldown; surface or dielectric loss to Q, field participation, and power sweeps. A material-only selection would appear in the control resonator, and a sensitivity-bounded or nuisance-degenerate outcome would place the shell coefficient below the recovery gate or close to a measured loss-channel template.

Perez, de Jong, Xu, and Gurevich supply the branch basis: condensation-energy NEMS concepts, superconducting drum force readout, cryogenic plate sensing across a superconducting transition, and microwave-loss reviews covering quasiparticles, residual resistance, dielectric loss, trapped vortices, kinetic inductance, and high-field limits [12, 17, 18, 16]. Those mechanisms are the nuisance family in Eq. 5.

The screening ladder gives superconducting rows from S/B = 3.59 to S/B = 17.76 across the viable nominal and optimistic cases. The apparatus includes shell-coupled and material-only resonators on the same cooldown path. Temperature sweeps across and below T_{c}, magnetic-history tests, vortex-loading controls, drive-power sweeps, quasiparticle checks, and geometry modulation all become evidence-bearing coordinates.

9. CONTROL MATRIX AND OUTCOME CLASSES

The control matrix distinguishes shell selection from material schedule selection, resonator dynamics, sensitivity closure, and central-reference closure. Table 3 summarizes the control standard for each physical branch.

Table 3: Evidence-bearing controls for physical readout branches.

Branch	Required controls	Shell-coherent data signature
Differential pressure	Matched dummy cell, opposite gap schedule, Kelvin-probe map, roughness prior, membrane drift model, thermal and vibration monitors	Pressure vector selects the shell schedule after patch, roughness, and membrane templates are included
High-Q cavity	Eigenmode model, shell-overlap and low-overlap modes, dummy cavity, thermal expansion monitor, loss and port-coupling sweeps	Frequency or phase vector selects shell overlap over loss, thermal, waveguide, and mode-mixing templates
Superconducting resonator	Material-only resonator, temperature sweep, drive-power sweep, magnetic-history control, quasiparticle checks, surface-loss model	Microwave response selects shell schedule after kinetic-inductance, loss, vortex, quasiparticle, and material templates are included

Table 4: Outcome classes for shell-geometry measurements.

Outcome	Data signature	Physical assignment
Shell coherent	Shell coefficient exceeds recovery gate and controls satisfy the false-positive target	The White sphere–cylinder shell schedule appears as a measurable Casimir/Lifshitz boundary response in that transducer
Material selected	Patch, capacitance, impedance, contact, or loss template has the stable coefficient	Ordinary material or electromagnetic response carries or masks the White shell schedule
Resonator selected	Cavity or superconducting response follows mode mixing, conductor loss, surface loss, vortex, or quasiparticle template	Resonator material dynamics explain the observed modulation under the tested settings
Sensitivity bounded	Shell and nuisance templates remain below recovery gate after calibration	The branch sets an upper bound on transducible White-shell response
Central reference bounded	Central propagation/timing coefficient remains far below the recovery gate	White-style central transit readout remains below laboratory recovery scale in this audit class

10. IMPLICATIONS

A shell-coherent result would promote the reported sphere–cylinder morphology from a worldline-numerics shell-like pattern to a measurable finite Casimir/Lifshitz boundary-response template in the annular sphere–wall region. The metric comparison would supply scale context, while central timing would remain an amplitude-bounded reference channel under the current audit.

Pressure recovery would give the annular morphology a direct mechanical stress projection. High-Q cavity recovery would give it an electromagnetic eigenmode projection through mode overlap. Superconducting recovery would give it a cryogenic microwave projection after kinetic-inductance, quasiparticle, vortex, surface-loss, magnetic-history, and material controls.

Cross-branch coherence would be the strongest result: recovery of a common shell coefficient in pressure, high-Q cavity, and superconducting readouts under independent nuisance ledgers. That pattern would identify the annular sphere–wall geometry as a shared finite-boundary response source and would favor a shared boundary source over branch-local pressure, cavity-mode, superconducting-loss, or material/patch explanations.

Pressure-only recovery would point to a mechanically transducible stress response with weak resonant overlap in the tested designs. Cavity-only recovery would point to eigenmode overlap as the cleaner transduction route. Superconducting-only recovery would point to a cryogenic microwave projection and would place special weight on material-only resonators, magnetic-history controls, temperature sweeps, and drive-power sweeps.

Material-, patch-, capacitance-, or loss-selected results would assign the White schedule to ordinary apparatus physics in that implementation. Sensitivity-bounded results would set upper limits on pressure, cavity, or superconducting transduction of the White shell coefficient. Central timing closure would assign the direct transit/timing interpretation to the bounded reference channel under the modeled backgrounds.

For finite-boundary Casimir metrology, engineered Casimir geometry would become a template-recovery problem across real precision instruments. The follow-up would extend the field beyond standard plate/sphere force laws into geometry-coded stress, eigenmode, and cryogenic microwave readouts. A shell-positive result would show that a designed finite boundary can organize a measurable response across one or more field-native transducers; bounded or background-selected results would define apparatus limits on measurable shell transduction.

11. REPRODUCIBILITY AND SOURCE BASIS

The citable scientific basis for this manuscript is the public literature in the reference list. The references provide the canonical DOI, arXiv, NIST, and journal locations for the source material.

The numerical screening values quoted in Tables 2 and 3 come from a finite readout-ladder calculation using shell-template recovery, electromagnetic-only false-positive checks, and physical backend cases for pressure, cavity, superconducting, material, and central-reference channels. The associated supplementary material consists of compact machine-readable screening summaries: candidate definitions, backend physical cases, gate summaries, and run-level summary metadata. Archiving the manuscript source and PDF with those summaries keeps the quoted design-screening values traceable.

12. CONCLUSION

The White sphere–cylinder geometry has an experimental role map: the sphere and inner wall set the finite boundary condition, the annular gap carries the shell template, material response is calibration, and the central current, photon, or electron transit comparison is an amplitude-bounded reference. The remaining test is branch-native transduction.

In this follow-on setting, readout names the full measurement ensemble: transducer, observable, controls, and background ledger. Pressure, cavity, and superconducting ensembles provide the three branch-native tests: direct stress, electromagnetic eigenmode perturbation, and cryogenic microwave boundary response. Shell coherence assigns the annular sphere–wall morphology to a measurable finite-boundary response; material or resonator selection assigns the same schedule to ordinary apparatus physics; sensitivity bounds set upper limits on measurable White-shell transduction. The original visual metric comparison becomes a field-native boundary-measurement problem whose next result identifies where the White shell coefficient lands.

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